

T5Gemma 2 vs. Qwen 2.5 on TReB: Interim Report

What we learned

akumch @ DiffusionTableQA

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1 What we learned

Upstream bug caps T5Gemma 2 at 4K input tokens, not 128K. At `batch_size=1`, eager attention, inputs > 4094 tokens raise `RuntimeError: size of tensor a (4097) must match tensor b (N)`. Constant $4097 = 4 \times \text{sliding_window}(1024) + 1$; $N = \text{input_length} + 3$. `sdpa` same class; `flash_attention_2` explicitly unsupported. Filed [#45521](#); FA2 feature request filed [#45522](#).

input tokens	result	tensor b dim
2,497, 3,525	OK	—
5,015	FAIL	5,018
6,499	FAIL	6,502
7,493	FAIL	7,496
9,997	FAIL	10,000
14,967, 19,808, 25,135	FAIL	input + 3

Base T5Gemma 2 never learned to stop. On plain “Q: 2+2? A:” it outputs 4 (correct), then loops Q: 2+2? A: 4 until `max_new_tokens`. EOS is not reliably emitted. On TCoT-format prompts it outputs nothing at all — the instruction format is OOD for this pretrained checkpoint (Google didn’t release an IT variant). This is a pretrained-model property, not something our SFT introduced.

Our FLAN-SFT adapter traded content for format. Training loss $3.22 \rightarrow 1.47$ over 782 steps on 100K FLAN samples — real learning, but mediocre plateau (smoke-test overfit on 200 samples got to 0.32). At inference: always emits one {“answer” : N } JSON then loops.

Prompt	Gold	Base output	SFT output
Q: 2+2? A:	4	4 then Q/A loop	$22+22=44 \times 100$
Math TCoT (Molly’s cost)	\$101	empty (0 tokens)	{“answer”: 100} + loop
Robustness TCoT	neutral	empty / garbled	{“answer”: 10} + loop
Code_Gen TCoT	Python code	empty	{“answer”: 1..125}
Plain total-cost question	\$101	—	\$1.15 and \$1.10 $\times 100$

So SFT gained: always commits to JSON envelope. SFT lost nothing structurally (EOS was already broken). What SFT never gained: content understanding on anything beyond single-scalar answers. Not mode-collapse in the “weights are random” sense — the number 100 for a \$101 problem shows task-weak correlation — but mode-collapse in the “learned shape, not semantics” sense.

Qwen 2.5 7B Instruct works end-to-end via vLLM. 7,789 TReB English samples, full 32K context, ~ 7 min wall-clock, $\sim \$1$ on $1 \times \text{H100 NVL}$.

Metric (overall, 7,789)	Value	Best task	Worst task
Exact Match	0.159	Table_Fact_Checking 0.914	Hallucination_Eval 0.000
ROUGE-L	0.309	Table_Fact_Checking 0.914	Table_Column_Naming 0.028
Numeric Match	0.434	—	—

Reading these requires care: spot-checks show Qwen produces multi-paragraph step-by-step reasoning but rarely wraps the final answer in `{"answer": ...}`. Tasks with EM=0 and ROUGE=0.3–0.5 (Understanding, Title_Naming, Summary) are format-not-capability failures. The judge’s 12-class failure-mode taxonomy will disentangle these; judge run deferred until the SFT is usable.

Pipeline hardening shipped during this session:

- Incremental `.partial.jsonl` writes with `fsync` every 5 batches — saved 95/100 after a crash where previously we lost everything.
- Token-budgeted prompt assembly (`build_prompt_tcot`): guarantees question + format-instruction survive. Previously right-truncation silently dropped the question on 12% of samples ($\geq 2K$ tokens) — 1-in-8 old predictions were answered without the model seeing the question.
- `merge_adapter.py` auto-infers base from `adapter_config.json` (prevents `PeftModel.from_pretrained` silent-corruption when the hand-typed base doesn’t match training).
- Sentinel (`predictions.jsonl.done`) + `monitor_pod.sh`: on sentinel detection, `scp` → verify line count matches sentinel → DELETE pod. No sentinel = pod stays alive for diagnosis.

2 What’s next

T5Gemma needs a re-SFT to be useful for this comparison. Priority fixes, in order:

1. **Fix EOS.** Audit FLAN tokenization — append a proper EOS token to every training target. This is likely why both base and SFT loop. Highest leverage.
2. **Use reasoning data.** FLAN targets are mostly short scalar answers; model never learned to reason. Train on FLAN-CoT / StrategyQA / GSM8K with full reasoning traces, not just final answers.
3. **More epochs.** Eval loss 1.38 at 782 steps is undertrained.

Without waiting for a new SFT, we can run Qwen@4K (client-side truncated) against T5Gemma base on the 100 stratified samples. Won’t answer “does SFT help encoder-decoders?” but will answer “does architecture matter pre-IT?” — honest, narrower result.

Upstream dependency. Both transformers issues need to land before T5Gemma 2’s advertised 128K context is reachable via HF. Related Gemma-family issues historically resolved on 1–6 week timelines.

Pointers

Project repo	https://github.com/junos-ai-org/jiutian-treb (branch <code>experiment-setup</code>)
Upstream bug	https://github.com/huggingface/transformers/issues/45521
Upstream FA2 request	https://github.com/huggingface/transformers/issues/45522
Bug repro (public)	<code>insights/transformers_bug_repro.py</code>
Eval driver / REPL	<code>eval.py</code> , <code>play.py</code> , <code>monitor_pod.sh</code>
SFT adapter	Hub DiffusionTableQA/t5gemma-2-4b-flan-sft-ddp, commit <code>abae90a8</code>
Qwen predictions (7,789)	<code>results/qwen_7b_instruct/predictions.jsonl</code>
T5Gemma base (100)	<code>results/t5gemma_base/predictions.jsonl</code>